Membrane protein diffusion sets the speed of rod phototransduction

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Retinal rods signal the activation of a single receptor molecule by a photon[1]. To ensure efficient photon capture, rods maintain about 10^9 copies of rhodopsin densely packed into membranous disks[2]. But a high packing density of rhodopsin may impede other steps in phototransduction that take place on the disk membrane[3], by restricting the lateral movement of, and hence the rate of encounters between, the molecules involved[4–6]. Although it has been suggested that lateral diffusion of proteins on the membrane sets the rate of onset of the photoresponse[7], it was later argued that the subsequent processing of the complexes was the main determinant of this rate[8,9]. The effects of protein density on response shut-off have not been reported. Here we show that a roughly 50% reduction in protein crowding achieved by the hemizygous knockout of rhodopsin in transgenic mice accelerates the rising phases and recoveries of flash responses by about 1.7-fold *in vivo*. Thus, in rods the rates of both response onset and recovery are set by the diffusional encounter frequency between proteins on the disk membrane.

Phototransduction begins when photoexcited rhodopsin (R*) and the G protein transducin (G) diffuse on the disk membrane, collide and then bind (reviewed in ref. 2). In this complex, R* catalyses the exchange of a GTP for a GDP on the Gα subunit and then the activated Gα (G*) is released. This process repeats many times for each R*, and constitutes the first amplifying step in the phototransduction cascade. G* diffuses along the membrane surface eventually activating a single cyclic GMP phosphodiesterase (PDE) subunit. Hydrolysis of cGMP by G*-activated PDE results in the closure of cationic channels in the plasma membrane and the hyperpolarizing photoresponse. Photoresponse recovery similarly requires the diffusional contact between proteins on the membrane (reviewed in ref. 10). R* shut-off begins with the phosphorylation of rhodopsin by the membrane-associated rhodopsin kinase and is complete after the binding of a soluble arrestin molecule. The timely shut-off of G*–PDE depends on its interaction with a membrane-associated regulator of G-protein signalling, RGS9, which accelerates the hydrolysis of Gα-bound GTP. Finally, cGMP synthesized by guanylate cyclase reopens the cationic channels.

To identify which process—diffusional encounter between components or molecular processing—limits the speed of phototransduction, we accelerated the frequency of protein encounters in intact photoreceptors and looked for changes in photoresponse kinetics. The frequency of encounters between cascade components, ν_{enc}, is proportional to the sum of their diffusion coefficients[11]. For a single R* and transducins[2],

$$\nu_{\mathrm{enc}} = 1.5(D_{\mathrm{R}} + D_{\mathrm{G}})C_{\mathrm{G}} \qquad (1)$$

where D_{R} and D_{G} are the diffusion coefficients of R* and G, respectively and C_{G} is the packing density of transducin in the disk membrane. The diffusion coefficients are sensitive to molecular crowding, which reduces the mobilities of the proteins by decreasing their mean free paths of diffusion. This has been demonstrated by Monte Carlo simulations[3,12] and by experimental measurements of the effects of protein concentrations on lateral diffusion[4–6].

For example, in the case of bacteriorhodopsin, an integral membrane protein whose cross-sectional area is similar to that of rhodopsin, the diffusion coefficient is inversely proportional to the protein packing density for molar phospholipid to protein ratios between 210 and 30 (ref. 4). In rod disk membranes, the phospholipid to rhodopsin ratio is about 65, and the diffusion coefficient for rhodopsin[13,14] is consistent with the relationship described for bacteriorhodopsin. Thus, in the disk membrane, where diffusion of R* and transducins is restricted by an enormous number of quiescent rhodopsin molecules, ν_{enc} will increase with the removal of some of the rhodopsin molecules. This has been achieved by the deletion of one of the rhodopsin alleles in transgenic mice[15].

Microspectrophotometric measurements showed that individual $\mathrm{R}^{+/-}$ rods contained about half of the normal amount of rhodopsin[15]. Other vital transduction components including transducin[15], PDE[15], RGS9, rhodopsin kinase, recoverin and arrestin (Fig. 1) were unchanged. The geometry of the outer segments and the spacing of the disks were also unchanged, suggesting that phospholipid molecules replaced the missing rhodopsins. We further support this conclusion with measurements of the phospholipid to rhodopsin ratio in purified disk membranes from $\mathrm{R}^{+/-}$ and wild-type rods (Table 1). Considering a density of 25,000–27,000 rhodopsins $\mu\mathrm{m}^{-2}$ in wild-type disk membranes[2] and a rhodopsin cross-sectional area of $\sim 10\,\mathrm{nm}^2$ (ref. 16), rhodopsin normally occupies ~27% of the disk membrane surface area. The 2.7-fold increase in the phospholipid to rhodopsin ratio in $\mathrm{R}^{+/-}$ rods

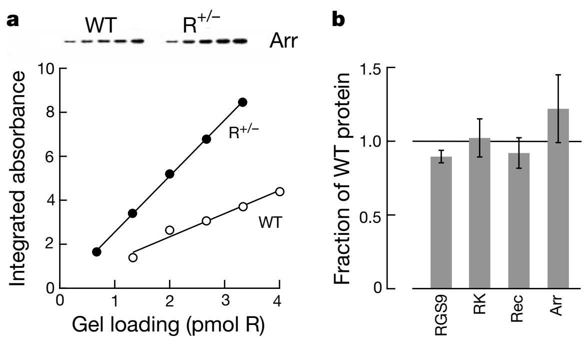

Figure 1 Normal levels of other proteins in $\mathrm{R}^{+/-}$ rod outer segments. **a**, Normal arrestin levels in $\mathrm{R}^{+/-}$ rods. Top, protein bands detected by ECL. Bottom, integrated optical density is proportional to the amount of rhodopsin loaded. Linear regression yielded slopes of 1.1 and 2.5 for wild type (WT; open circles) and $\mathrm{R}^{+/-}$ (filled circles), respectively, which are consistent with a roughly 50% decrease in rhodopsin levels in $\mathrm{R}^{+/-}$ rods without a change in arrestin levels. **b**, Mean ± s.e.m. $\mathrm{R}^{+/-}$ protein levels relative to wild type, obtained from at least three determinations for each protein. Arr, arrestin; Rec, recoverin; RK, rhodopsin kinase.

indicated that rhodopsin density in the disks decreased about 2.2-fold. The addition of extra phospholipids to $R^{+/-}$ disks did not perturb the high unsaturated fatty acid content appreciably (Table 1), so the intrinsic membrane fluidity would not be expected to change. But the relief of membrane crowding would increase the diffusion coefficients of R*, transducin and other membrane proteins roughly twofold[3,4]. ν_{enc} and the rates of diffusion-limited reactions taking place on the membrane surface would increase proportionately.

For equivalent numbers of absorbed photons, flash responses in $R^{+/-}$ rods developed faster and peaked sooner than in wild-type rods, but their amplitudes were virtually equal (Fig. 2a, Table 1). We estimated the degree to which the onsets of responses to flashes suppressing less than 50% of the circulating current were accelerated in $R^{+/-}$ rods by scaling the initial rising phases of wild-type responses to coincide with those of $R^{+/-}$ (see Fig. 2b). After averaging the scaling factors for the three dimmest flash strengths, the response onset was 1.67 ± 0.06 times faster in $R^{+/-}$ rods.

Bright flash responses were fitted with a quantitative model relating the slope of the rising phase to the degree of cascade amplification[2]:

$$\frac{r(t)}{r_{max}} = 1 - \exp\left[-\frac{1}{2}\phi A(t - t_{eff})^2\right] \qquad (2)$$

where $r(t)$ is the response amplitude at time t after the flash, r_{max} is the maximal response, ϕ is the number of R*, t_{eff} is an intrinsic delay in the response onset and A is the amplification constant. Equation (2) explicitly neglects inactivation processes that substantially

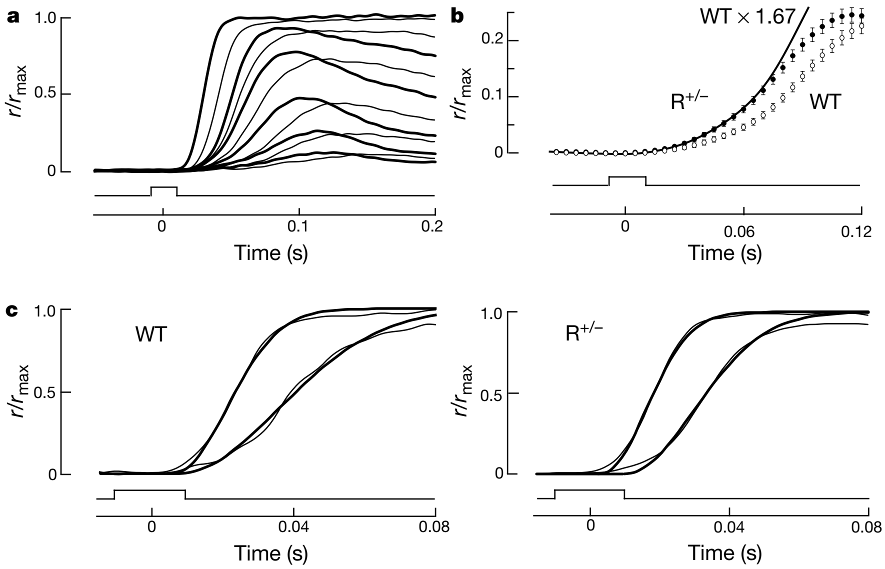

Figure 2 Accelerated flash response onset in $R^{+/-}$ rods. Lowest traces indicate the flash monitors. Numbers of R* were based on photon-collecting areas of 0.47 and 0.23 μm^2 for wild type (WT) and $R^{+/-}$, respectively. **a**, Flash responses from $R^{+/-}$ (thick traces) and WT (thin traces) rods. Each trace is the average of at least ten responses from six WT or nine $R^{+/-}$ rods except where indicated in parentheses. The 500-nm flash strengths (mean ± s.d.) were WT, 10.7 ± 1.5, 22.1 ± 1.8 (n = 5), 40.4 ± 2.9, 96.0 ± 6.9, 172 ± 14, 1,196 ± 301 (n = 3); $R^{+/-}$, 23.7 ± 2.8, 44.5 ± 6.1 (n = 7), 101 ± 13, 192 ± 26, 404 ± 17 (n = 3), 2,790 ± 14 photons μm^{-2}. **b**, Responses of WT (open circles) and $R^{+/-}$ (filled circles) rods in **a** to flashes resulting in ~5.5 R*. Error bars indicated s.e.m. Solid line is WT response after scaling its amplitude by 1.67. **c**, Enhanced amplification constant in $R^{+/-}$ rods. Records were filtered at 80 Hz (see Methods). WT responses to 172 and 594 photons μm^{-2} (left) and $R^{+/-}$ responses to 400 and 1,400 photons μm^{-2} (right) were fitted with equation (2) (thick lines), yielding A values (in s^{-2}) of 15.2 and 16.3 (WT), and 23.4 and 22.4 ($R^{+/-}$).

Table 1 Flash response parameters and disk membrane composition

	Wild type	$R^{+/-}$	
Sensitivity, i_o (photons μm^{-2})*	60 ± 10 (14)	111 ± 10 (21)	$P < 0.001$
Dim flash response			
Time to peak (ms)*	140 ± 6 (10)	120 ± 2 (16)	$P < 0.01$
Integration time (ms)*	250 ± 18 (10)	180 ± 10 (16)	$P < 0.01$
Single-photon response amplitude (pA)	0.66 ± 0.08 (10)	0.64 ± 0.06 (10)	NS
Amplification constant, A (s^{-2})	13.6 ± 1.4	22.9 ± 1.5	$P < 0.01$
Saturating flash response			
τ_{c1} (ms)	200 ± 17 (10)	130 ± 7 (14)	$P < 0.001$
τ_{c2} (ms)	580 ± 92 (6)	350 ± 21 (11)	$P < 0.01$
$Na^+/Ca^{2+},K^+$ exchanger			
τ_{ex} (ms)	105 ± 13 (9)	125 ± 22 (11)	NS
Magnitude (% of suppressible current)	7.7 ± 1.1 (9)	8.1 ± 1.0 (11)	NS
Disk membranes			
Phospholipid/rhodopsin (mol mol^{-1})	86 ± 12 (11)	236 ± 18 (14)	$P < 0.001$
Mean no. double bonds per fatty acid	2.7 ± 0.2 (11)	2.3 ± 0.1 (14)	NS
Saturated/unsaturated fatty acids (mol mol^{-1})	0.95 ± 0.12 (11)	1.07 ± 0.07 (14)	NS
Fatty acid length	19.4 ± 0.1 (11)	19.0 ± 0.1 (14)	NS

Values given as mean ± s.e.m. (number of determinations). i_o is the half-saturating flash strength at 500 nm. Dim flash responses, whose amplitudes were less than a fifth of the maximum, had the same shape as the single-photon response and differed only in amplitude. Integration time was given by the integral of the response normalized by its amplitude. Amplitude of the single-photon response was found by dividing the response variance by the mean amplitude. A is the amplification constant from equation (2). The average A values were obtained from responses to flashes delivering 172, 594, 1,196 and 2,660 photons μm^{-2} (wild type) and 192, 404, 775, 1,400 photons μm^{-2} ($R^{+/-}$) as described in Fig. 2. τ_{c1} is the dominant time constant of the recovery for saturating flashes of < 2,000 photons μm^{-2} and τ_{c2} is that for saturating flashes of > 2,000 photons μm^{-2}. τ_{ex} is the time constant for $Na^+/Ca^{2+},K^+$ exchange. NS, not significant.
* Includes results from ref. 15.

influence the earliest detectable rise of responses to weak flashes but it is applicable to bright flash responses because their rising phases develop much more quickly than inactivation (Fig. 2c). In $R^{+/-}$ rods A was 1.7-fold higher than in wild-type rods (Table 1), in good agreement with the scaling factor found for dim flashes.

The amplification constant may be expressed in the following way[2]:

$$A = \nu_{RG} p_{GP} \beta n \tag{3}$$

where ν_{RG} is the rate of transducin activation by a single R*, p_{GP} is the coupling efficiency between activated transducin and PDE, β is the hydrolytic rate constant of each PDE subunit and n is the Hill coefficient for activation of the cGMP-gated channel. As $p_{GP,}$ β or n are not expected to have been altered by the deletion of a single rhodopsin allele, we attribute the entire 1.7-fold increase in A to a corresponding increase in ν_{RG}.

The rate of the multi-step activation of transducin may be expressed as:

$$\nu_{RG} = \frac{1}{\tau_{enc}/p_{RG} + \tau_{proc}} = \frac{1}{1/(\nu_{enc} p_{RG}) + \tau_{proc}} \tag{4}$$

Here $\tau_{enc} = 1/\nu_{enc}$ is the average time between R* and G·GDP collisions, p_{RG} is the fraction of encounters that lead to formation of a complex, and the processing time τ_{proc} includes nucleotide exchange and dissociation of the complex. ν_{RG} must be lower than ν_{enc}, as p_{RG} is not unity and processing time is finite. Estimates of ν_{enc} range from 7,000 s^{-1} in amphibians to 17,000 s^{-1} in mammals[2], whereas ν_{RG} is thought to be 1.4–50 times lower[2,17]. The low values of ν_{RG} were attributed to a long τ_{proc} on the assumption that R*–G coupling was highly efficient[2]. However, in $R^{+/-}$ rods there is no reason to believe that the reduction of R packing density might shorten τ_{proc} of G activation. It is more likely that ν_{RG} increased because $1/(\nu_{enc} p_{RG})$ was slower than τ_{proc}. Then it is the frequency of productive encounters between R* and G that sets the rate of photoresponse onset. Notably, the low rate of transducin activation reported[17] infers that the fraction of effective collisions between R* and G is 0.01–0.1, which is substantially lower than previously supposed.

Photoresponses also recovered more rapidly in $R^{+/-}$ rods. For all flash intensities studied, the substantially faster falling phase in the $R^{+/-}$ rods shortened the times to peak and abbreviated response integration times (Fig. 3a, b; Table 1). Responses to bright flashes recovered in two phases. Only the faster phase was accelerated in $R^{+/-}$ rods. The slower phase, which probably arises from the failure of a small fraction of photoexcited rhodopsins to be phosphorylated by rhodopsin kinase[18], was similar in $R^{+/-}$ and wild-type rods.

One possible explanation for accelerated response recovery in $R^{+/-}$ rods might be that the Ca^{2+} feedback systems operated more quickly. During the normal photoresponse the recovery is accelerated because of Ca^{2+} feedback[10]. Ca^{2+} enters the outer segment through the cGMP-gated channel and is extruded by a Na^{+}/Ca^{2+}, K^{+} exchanger. Closure of the channels during the photoresponse blocks the entry of Ca^{2+}, but its efflux continues so intracellular Ca^{2+} declines. This decline is thought to relieve an inhibition of rhodopsin kinase by Ca^{2+}-recoverin, causing R* to be shut-off faster, and to stimulate cGMP production through guanylate-cyclase-activating proteins.

Key factors that set the dynamic behaviour of Ca^{2+} feedback include the amplitude and rate of Ca^{2+} decline in response to light and the Ca^{2+} dependence of the feedback target[19]. Figure 4 shows that the amplitude and rate of light-stimulated Ca^{2+} decline were not different in wild-type and $R^{+/-}$ rods and that the Ca^{2+} dependence of cGMP synthesis, the most potent site of Ca^{2+} feedback in rods[20], was normal in $R^{+/-}$ rods (see also Table 1). Furthermore, the rhodopsin kinase and recoverin levels were normal (Fig. 1). Thus, Ca^{2+} feedback systems do not seem to be changed in $R^{+/-}$ rods.

Another explanation is that the cascade shuts off sooner owing to a reduction in protein crowding that accelerates protein–protein interactions. The shut-off of the phototransduction cascade has been studied using the Pepperberg method[21,22]. Stimulation of rods with very bright flashes caused them to remain in saturation, with all cGMP-gated channels closed, for a period of time that depended on the flash strength. In some species the time in saturation was linearly related to the natural logarithm of flash intensity, where the slope gave the time constant of the slowest cascade shut-off step[21]. In mouse rods this relationship was not linear, indicating that several, progressively slower processes may limit the shut-off for flashes of increasing intensities (Fig. 3c). Nevertheless, the relationship between the time in saturation and the number of absorbed photons was less steep in the $R^{+/-}$ rods than in the wild-type rods. For comparison, the relations were approximated by linear regression of the data over two intensity domains. In each domain the slope for

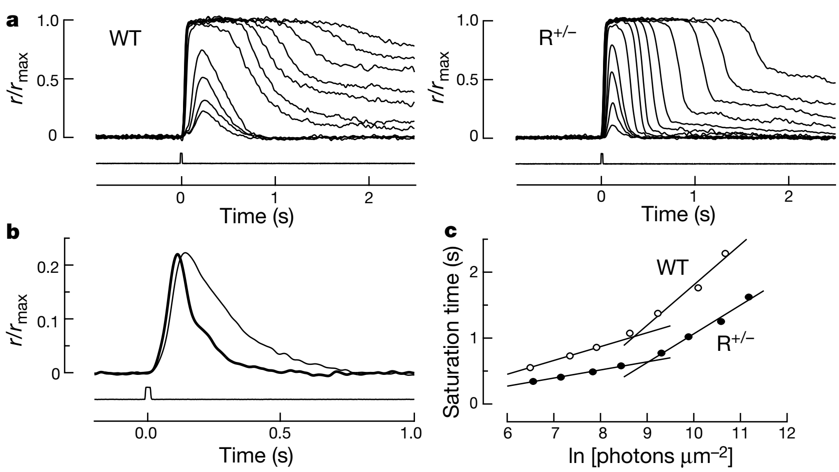

Figure 3 Accelerated flash response recovery in $R^{+/-}$ rods. **a**, Families of averaged flash responses from a wild-type (WT) and an $R^{+/-}$ rod. Flash strengths at 500 nm were (in photons μm^{-2}) WT, 6.4, 11.6, 23.2, 42.1, 659, 1,550, 2,810, 5,650, 10,200, 24,100 and 43,600; and $R^{+/-}$, 21, 38, 90, 164, 596, 1,400, 2,540, 5,100, 9,230, 21,800, 39,400, 79,200 and 143,000. The maximal response was 8.2 pA for WT and 11.3 pA for $R^{+/-}$. **b**, Averaged dim flash responses of five WT (thin trace) and seven $R^{+/-}$ (thick trace) rods to ~5.5 R*. **c**, Times that cells in **a** remained in saturation after bright flashes, measured from mid-flash until 20% recovery of the circulating current. The relationship for the $R^{+/-}$ rod was shifted left by 0.693 ln units to normalize for photon capture. Straight lines show linear regression analyses with slopes of 0.210 s for WT and 0.123 s for $R^{+/-}$ for flashes < 2,000 photons μm^{-2}, and 0.612 s for WT and 0.436 s for $R^{+/-}$ for flashes > 2,000 photons μm^{-2}.

$R^{+/-}$ rods was 1.5–1.7-fold less than that for the wild-type rods (Table 1). Thus, decreased membrane crowding also caused the dominant mechanism(s) controlling the cascade shut-off rate to proceed more rapidly.

On the one hand, the key rate-limiting recovery step may be rhodopsin quench, because in recoverin knockout mouse rods recovery was also accelerated[23]. In these rods, R* shut off sooner presumably because the higher concentration of Ca^{2+}-recoverin-free rhodopsin kinase increased the frequency of encounters between R* and rhodopsin kinase. On the other hand, G*–PDE shut-off has been implicated as the slowest recovery process in amphibian rods[22]. Other data[24] are consistent with the idea that formation of the complex between RGS9-Gβ5 and G* limits the rate of G* shut-off. So, the time required for phosphate transfer to R* by rhodopsin kinase or for RGS9-Gβ5 accelerated G* GTPase must be substantially shorter than the time between productive encounters of corresponding components. Increased rates of diffusion on the membrane will therefore cause faster quench of R* and/or G*.

In conclusion, lowering the rhodopsin concentration in rod photoreceptor membranes decreases photon capture, thereby reducing threshold sensitivity. At the same time the photoresponse speeds up, improving temporal resolution of the visual stimulus. Conversely, raising the rhodopsin packing density would increase light absorption, but would slow and eventually stop phototransduction owing to excessive protein crowding. Optimal solution of the problem may depend on the specific behavioural needs of the photoreceptor's owner. Furthermore, as the response speed is set by the frequency of protein–protein encounters, the photoreceptor might alter its sensitivity and temporal resolution to match long-term changes in ambient lighting by modulating the levels of certain transduction proteins. It will be interesting to see whether similar trade-offs exist between speed and sensitivity in other membrane-bound signal transduction systems. □

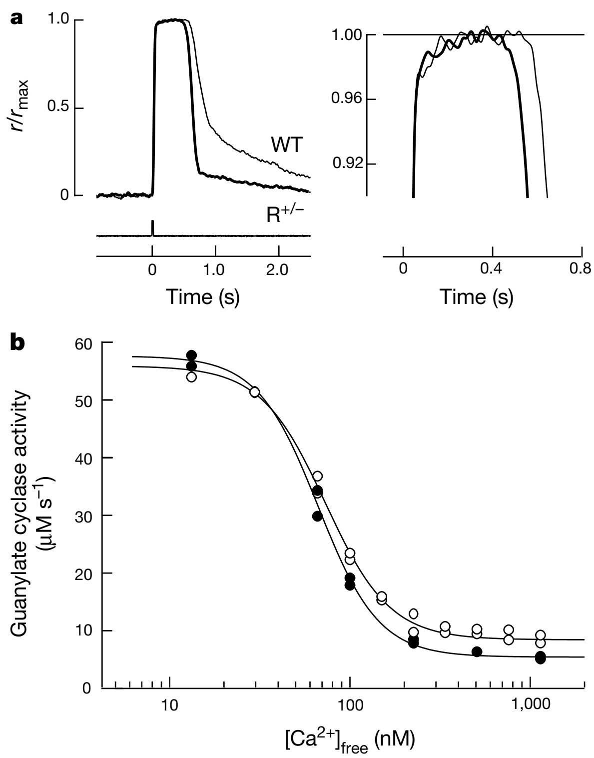

Figure 4 Ca^{2+} feedback was unperturbed in $R^{+/-}$ rods. **a**, Ca^{2+} dynamics in the rod outer segment. Left, saturating flashes prevented the entry of Ca^{2+} by closing all cGMP-gated channels in the plasma membrane. Right, the residual current at the peak of the response was due to removal of intracellular Ca^{2+} by the electrogenic $Na^+/Ca^{2+},K^+$ exchanger[30] (expanded scale). The similar amplitude and kinetics between wild-type (WT) (thin traces) and $R^{+/-}$ rods (thick traces) indicates that the intracellular Ca^{2+} content in darkness and the rate of light-stimulated Ca^{2+} removal were unchanged in $R^{+/-}$ rods. **b**, Normal Ca^{2+} dependence of cGMP synthesis in $R^{+/-}$ rods. Rate of cGMP synthesis is expressed as the change in concentration in the outer segment cytoplasmic volume over time ($\mu M\ s^{-1}$). Continuous lines show fits with the Hill equation: $\alpha(Ca) = \alpha_{max} - \Delta\alpha\ (Ca^n)/(Ca^n + k^n)$, where α is the rate of cGMP synthesis, α_{max} is the maximal rate, $\Delta\alpha$ is the difference between the highest and lowest rates, Ca is $[Ca^{2+}]_{free}$, k is a constant, n is the Hill coefficient. WT (open circles): $\alpha_{max} = 55.9\ \mu M\ s^{-1}$, $\Delta\alpha = 47.5\ \mu M\ s^{-1}$, $k = 73.0$ nM, $n = 2.4$. $R^{+/-}$ (filled circles): $\alpha_{max} = 57.6\ \mu M\ s^{-1}$, $\Delta\alpha = 52.2\ \mu M\ s^{-1}$, $k = 66.6$ nM, $n = 2.6$.

Methods

ROS preparation for biochemical analyses

Rod outer segements (ROSs) were collected from 8–10-week-old mice under infrared illumination[25]. Mice were dark adapted for a minimum of 12 h, anaesthetized with CO_2 and killed by cervical dislocation. Retinas were isolated into Locke's solution containing 6.7% Optiprep (Fisher), vortexed and large pieces were allowed to sediment. The supernatant was loaded onto a step gradient consisting of 6.7, 8.3 and 20% Optiprep. After centrifugation for 15 min at 1,425 *g* ROSs were collected from the 8.3–20% gradient interface. Optiprep was removed by several washings with Locke's solution containing 0.1 mM phenylmethylsulphonyl fluoride, 2 mM EDTA and 15 $\mu g\ ml^{-1}$ each of aprotinin, leupeptin and pepstatin. Rhodopsin content was determined spectrophotometrically.

Determination of relative ROS protein levels

ROS samples from $R^{+/+}$ and $R^{+/-}$ animals were dissolved in SDS sample buffer and specific protein levels were determined by western blot. Four to five samples containing increasing amounts of R from each mouse genotype were run in neighbouring lanes on a single 12.5% polyacrylamide gel, and transferred onto nitrocellulose membrane. We probed blots with primary antibodies as described[15] along with an additional anti-RGS9 antibody[26]. Horseradish peroxidase-conjugated secondary antibodies were used at a dilution of 1:10,000, visualized by enhanced chemiluminescence (ECL; Pierce), and proteins were quantified by scanning densitometry using an AlphaImager (Alpha Innotech, San Leandro, CA). Integrated optical densities of bands were plotted as a function of the amount of rhodopsin loaded. We used the linear region of this relationship to determine the amount of protein in $R^{+/-}$ ROSs relative to that in wild type.

Determination of guanylate cyclase activity

The assay of guanylate cyclase activity was carried out essentially as described[19]. The reaction was started by adding 5 μl of pseudo-intracellular medium containing GTP to 10 μl of ROS suspension. Final concentrations of reactants were (in mM): 0.02 ($R^{+/+}$) or 0.01 ($R^{+/-}$), 2 Mg^{2+}, 0.15 zaprinast, 3 [^{32}P]GTP·α, 30 cGMP, 0.06 ATP and the Ca^{2+}/BAPTA ratio required for the desired free Ca^{2+}. We quenched the reaction after 2 min by adding 60 μl 50 mM EDTA and boiling for 1 min. Samples were centrifuged for 10 min on a Beckman Microfuge E. Supernatant (5 μl) was loaded onto PEI-cellulose thin layer chromatography plates (EM Sciences, Gibbstown, NJ). Nucleotide separation was carried out with 0.2 mM LiCl. cGMP was visualized with ultraviolet light, cut from the plate, eluted with 2.0 M LiCl and counted in 10 ml of ScintiSafe cocktail.

Photoreceptor lipid content

Lipid analysis was carried out on photoreceptor disks prepared from ROSs purified as stated above. The disk preparation was a modification of the Ficoll floatation method developed for bovine disks[27]. We determined phospholipid content and fatty acid composition as described[28].

Single-cell suction electrode recording

Photoresponses were recorded as described[29]. We carried out all procedures under infrared illumination. Dark-adapted mice were anaesthetized with CO_2 and killed by cervical dislocation. Retinas were isolated into Leibovitz's L-15 medium (Gibco). Residual vitreous humor was removed and the tissue stored on ice. Small pieces of tissue were chopped finely, treated with DNase (type IV; Sigma), transferred to an experimental chamber and perfused with an enriched Locke's solution saturated with 95% O_2/5% CO_2. Locke's solution contained (in mM): 144 Na^+, 3.6 K^+, 1.2 Ca^{+2}, 2.4 Mg^{+2}, 123.3 Cl^-, 10 HEPES, 20 HCO_3^-, 0.02 EDTA, 10 glucose, 0.5 glutamate, 3 succinate, BME vitamins, MEM amino acids; pH 7.4. A rod outer segment was pulled into a glass suction electrode and the membrane currents monitored with a current-to-voltage recorder (Axopatch 200A, Axon Instruments, CA) at 36–38 °C. The solution inside the electrode did not contain vitamins or amino acids, and HCO_3^- was replaced with Cl^-. We stimulated rods with 21-ms flashes from a xenon arc source passing through a six-cavity interference filter that transmitted 500-nm light with a half-maximal transmission bandwidth of 10 nm (Omega Optical, Brattleboro, VT). Records were low-pass filtered at 30 Hz (−3 dB, 8-pole Bessel filter) and digitized at 400 Hz. The delay introduced by low-pass filtering was not removed from responses unless otherwise noted. Filtering at 30 Hz influenced the determination of the cell amplification constant from responses to bright flashes. Therefore, the filtering was digitally removed by deconvolution (Numerical Recipes) and then the records were low-pass filtered at 80 Hz.

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Translational repression determines a neuronal potential in *Drosophila* asymmetric cell division

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Asymmetric cell division is a fundamental strategy for generating cellular diversity during animal development[1]. Daughter cells manifest asymmetry in their differential gene expression. Transcriptional regulation of this process has been the focus of many studies, whereas cell-type-specific 'translational' regulation has been considered to have a more minor role. During sensory organ development in *Drosophila*, Notch signalling directs the asymmetry between neuronal and non-neuronal lineages[2], and a zinc-finger transcriptional repressor Tramtrack69 (TTK69) acts downstream of Notch as a determinant of non-neuronal identity[3,4]. Here we show that repression of TTK69 protein expression in the neuronal lineage occurs translationally rather than transcriptionally. This translational repression is achieved by a direct interaction between *cis*-acting sequences in the 3′ untranslated region of *ttk69* messenger RNA and its *trans*-acting repressor, the RNA-binding protein Musashi (MSI)[5]. Although *msi* can act downstream of *Notch*, Notch signalling does not affect MSI expression. Thus, Notch signalling is likely to regulate MSI activity rather than its expression. Our results define cell-type-specific translational control of *ttk69* by MSI as a downstream event of Notch signalling in asymmetric cell division.

Mechanosensory bristle development in *Drosophila* is an excellent model system in which to address the molecular mechanisms of asymmetric cell division[2]. Four successive asymmetric cell divisions from a common precursor cell called the sensory organ precursor (SOP) generate a sensory bristle comprising four different non-neuronal support cells and one neuron[6,7] (Fig. 1a, b and 2f, m). The first asymmetric cell division of SOP into IIa non-neuronal and IIb neuronal precursors is regulated by Notch signalling[2,8,9]; specific activation of Notch occurs in IIa owing to inhibition in IIb by Numb, an intracellular negative regulator of Notch[10]. Activation of Notch signalling results in the appearance of TTK69 protein in the IIa precursor, but not in IIb (refs 3, 10). The expression pattern of TTK69, phenotypes of *ttk69* loss-of-function mutants, and the overexpression of TTK69 (refs 3, 11) suggest that TTK69 is necessary and sufficient to specify the IIa non-neuronal lineage (Fig. 1c, d); however, the mechanism through which Notch signal-

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